

Blowing Candles

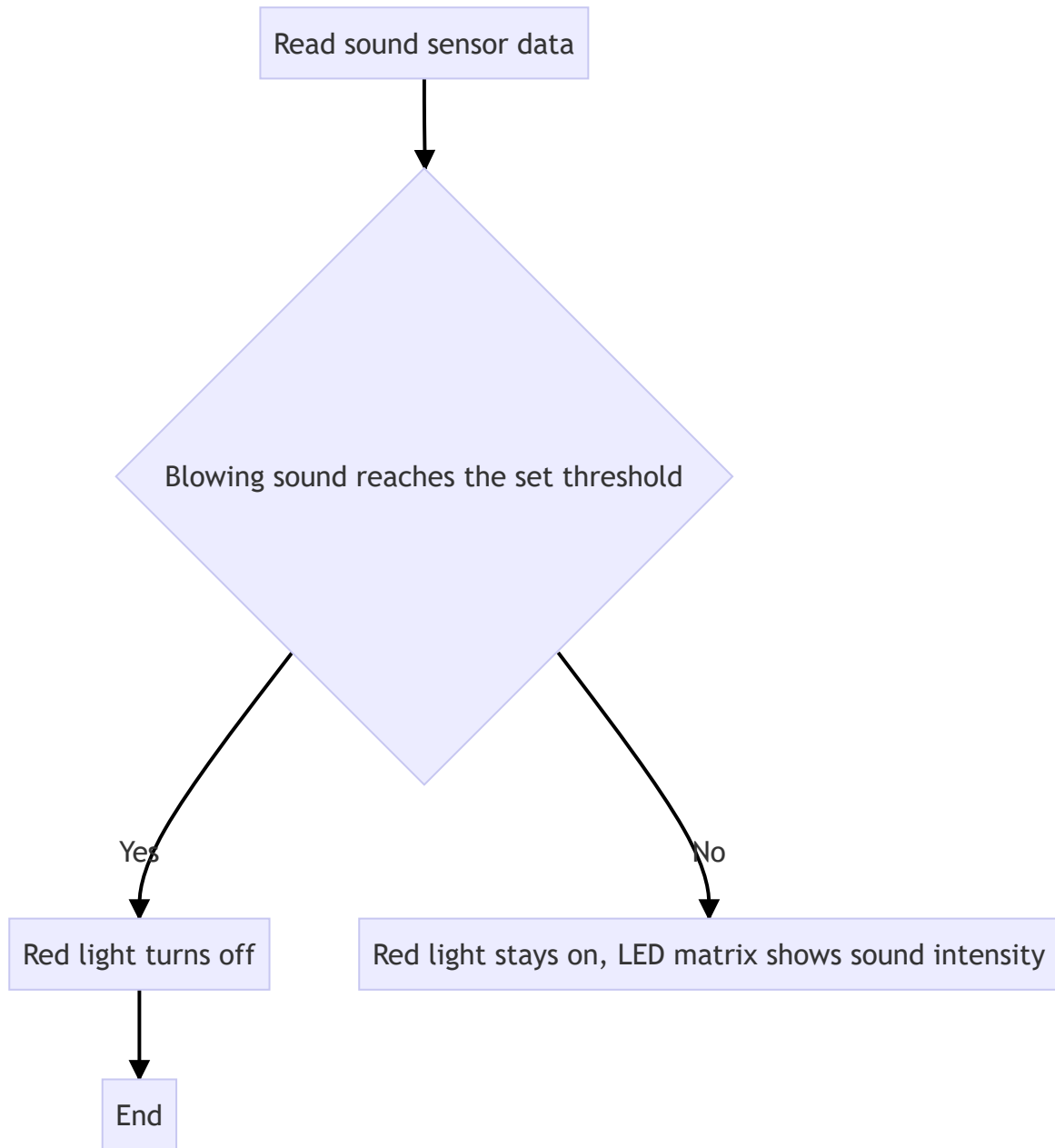
Blowing Candles

1. Experiment Purpose
2. Implementation Principle
3. Experiment Steps
4. Experiment Phenomenon

1. Experiment Purpose

In this section, we will learn to use the microphone sensor to detect sound intensity and achieve the "blowing candles" effect.

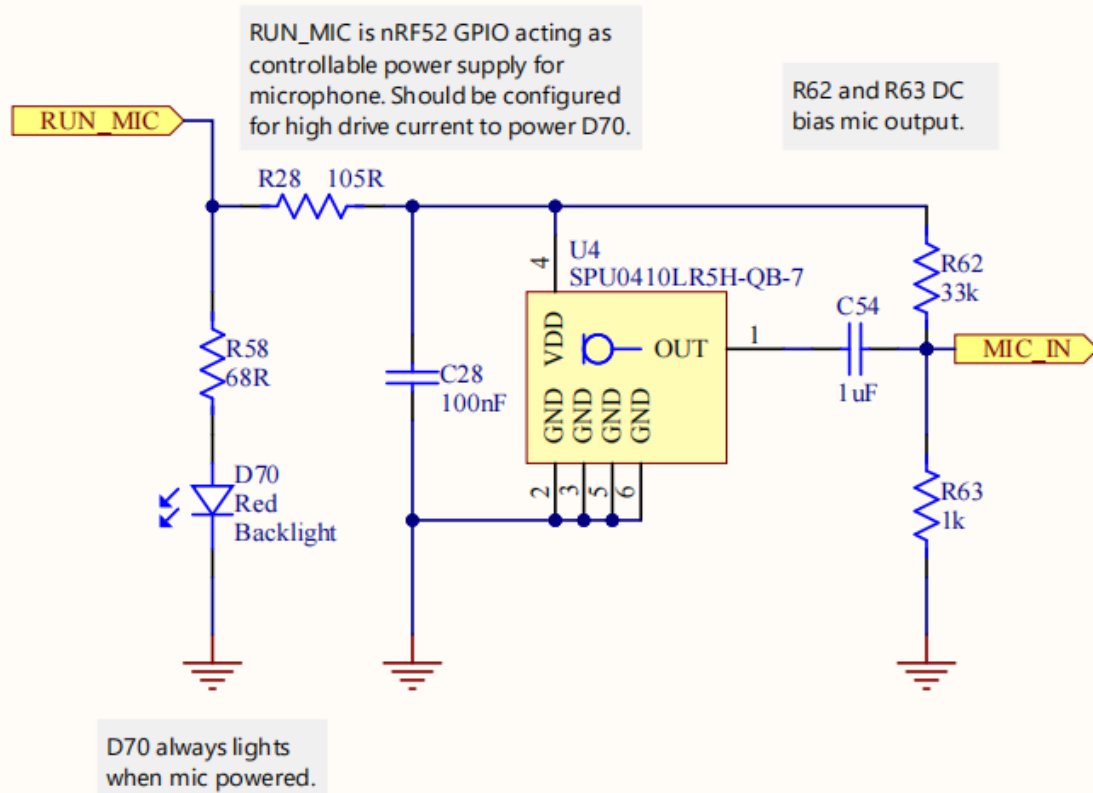
2. Implementation Principle



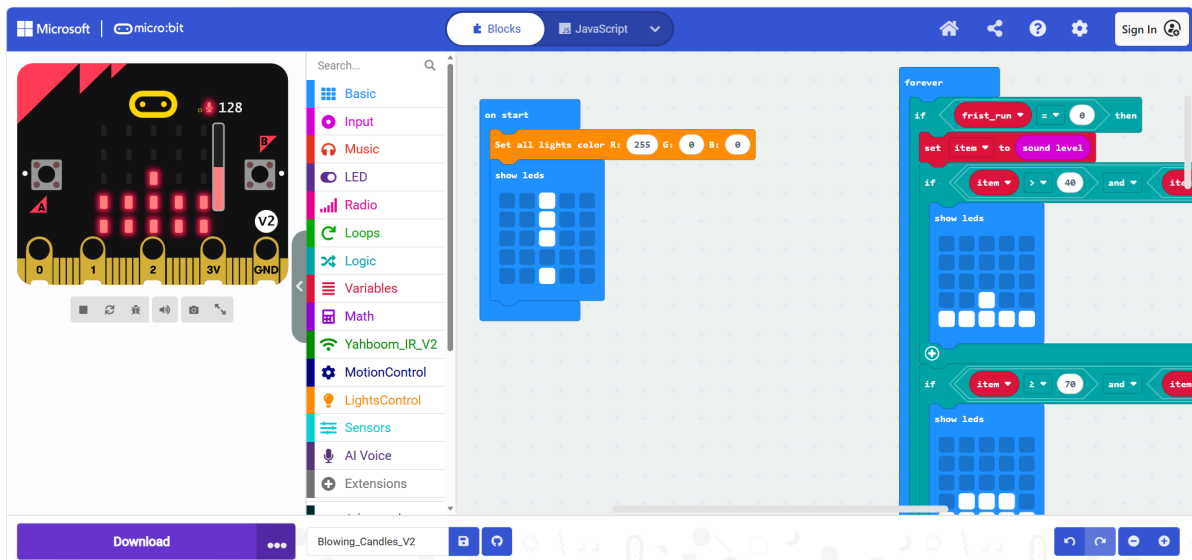
V2 version (Micro:bit V2 built-in microphone)

A microphone sensor is provided near the Logo of the Micro:bit V2. When the program drives and reads it, the red LED lights up; when it is not read, the red LED turns off and the microphone is shut down, which can reduce the power consumption of the Micro:bit at the hardware level.

Microphone



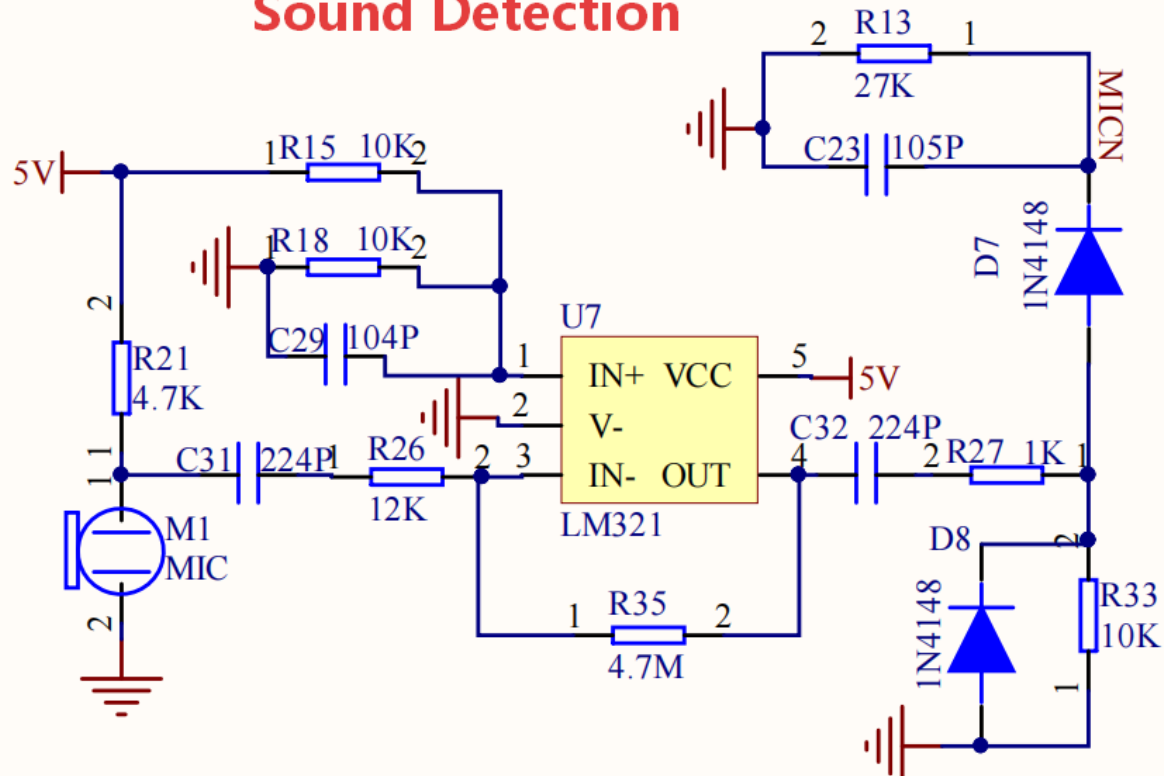
V2 version program



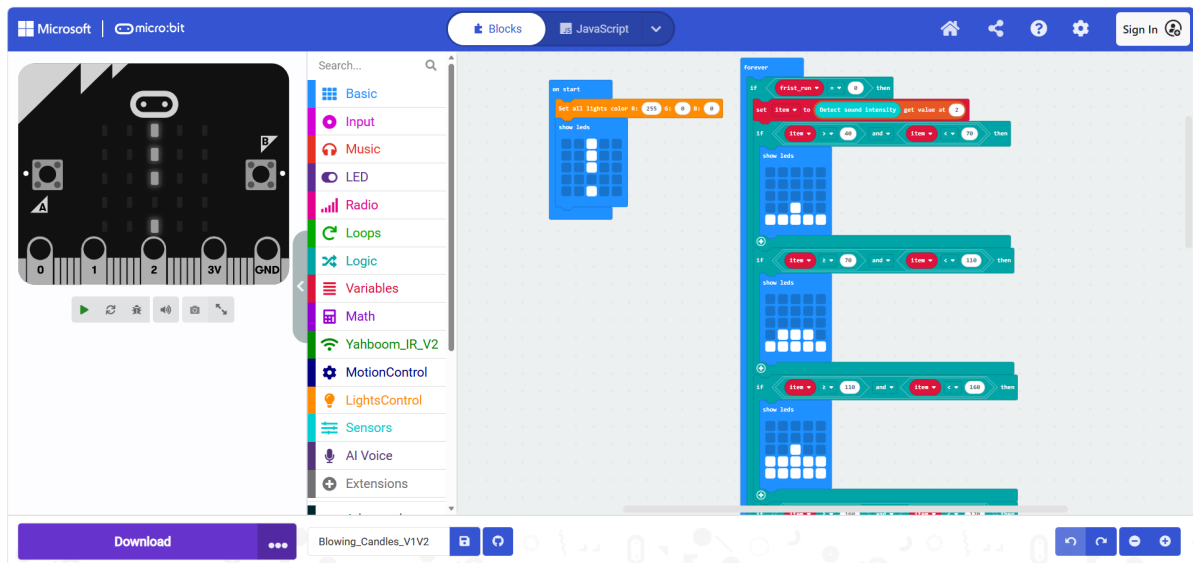
V1V2 version (expansion board MIC sensor)

The on-board sound sensor consists of an electret microphone (commonly called a mic head) plus signal amplification and clamping protection, forming a complete sound detection circuit.

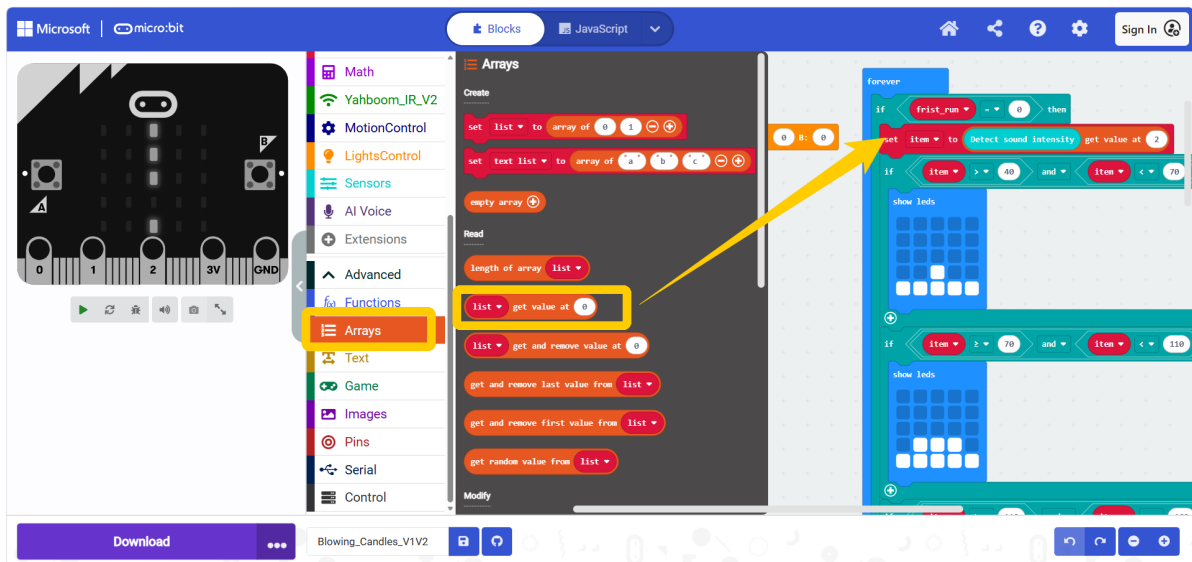
Sound Detection



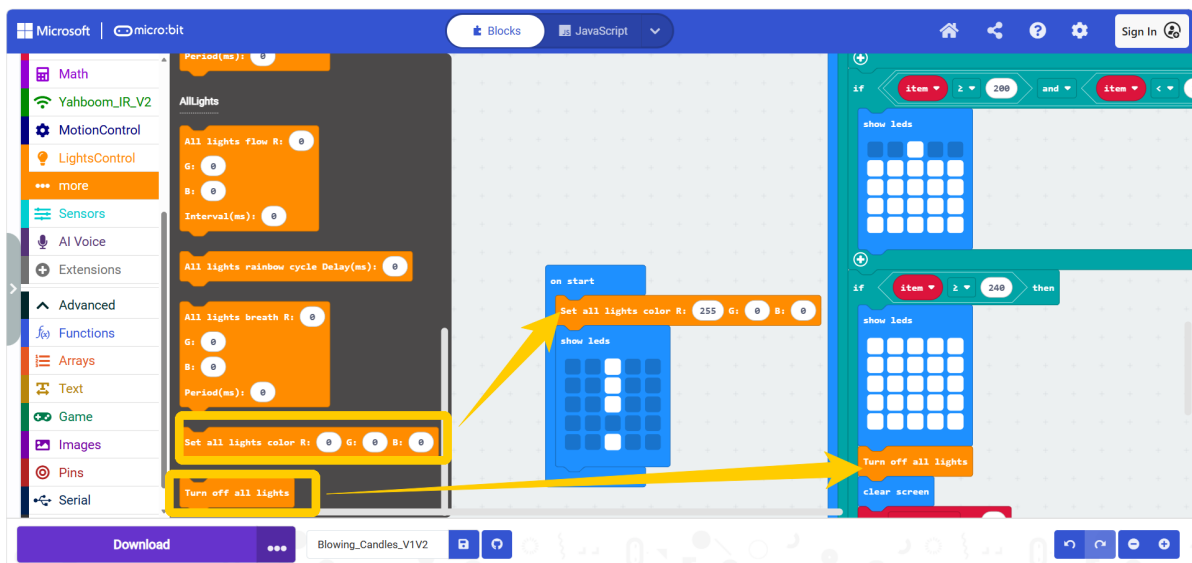
V1V2 version program



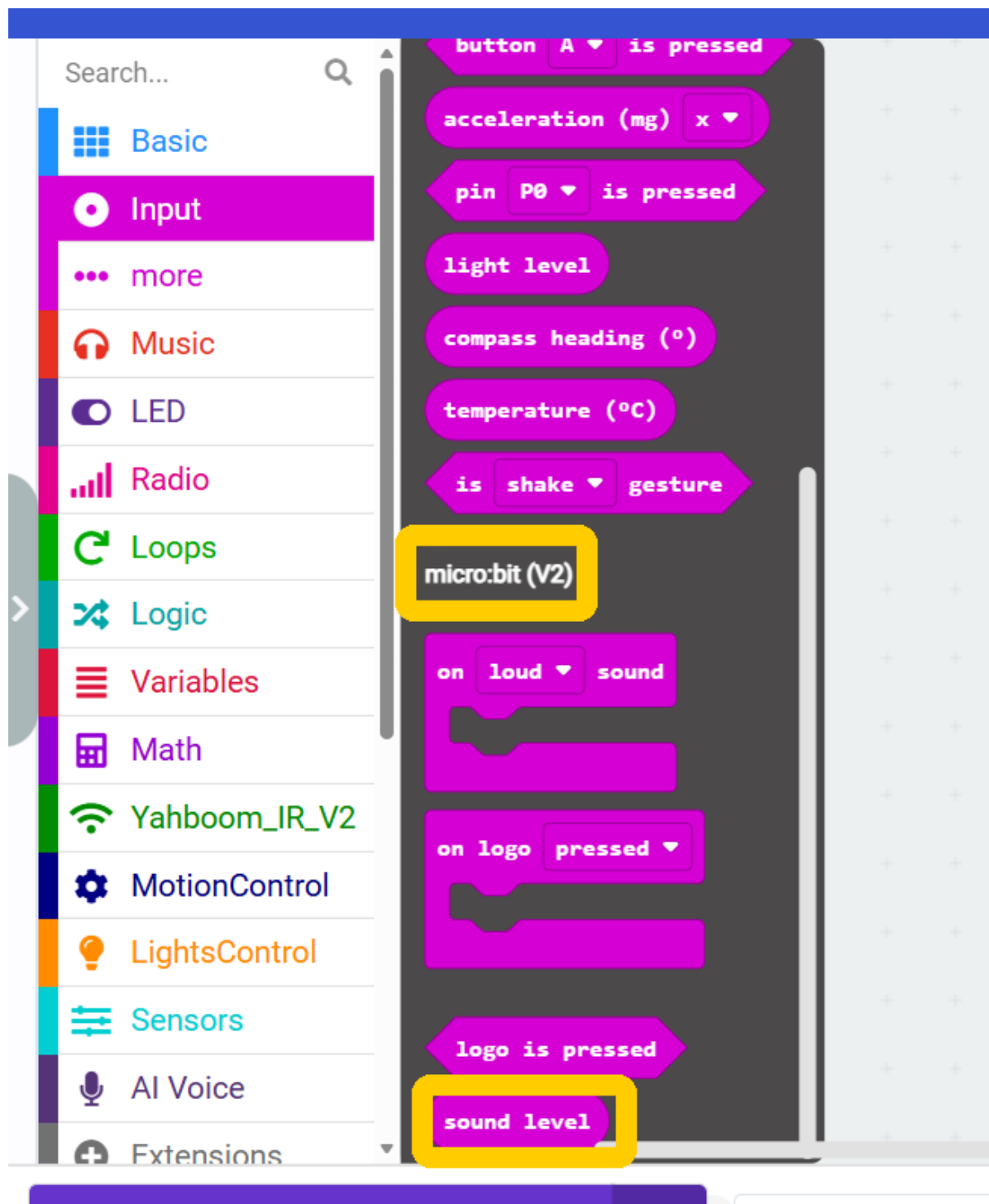
Reading the sound intensity index value from the array is located in the **Arrays** category.



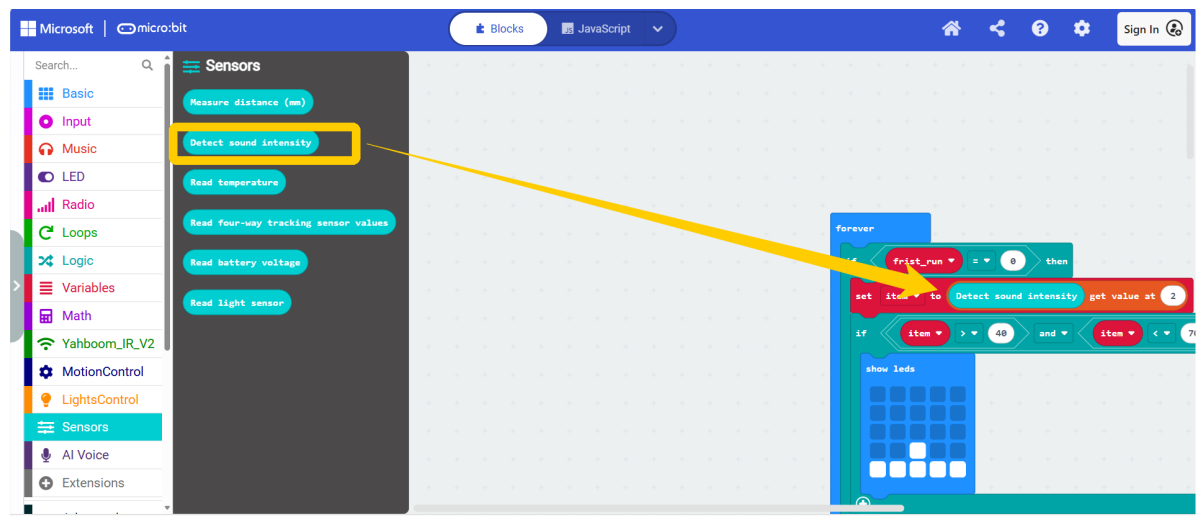
The light control part is located in the `LightsControl1` category.



The V2 version reads the sound loudness in the `microbit v2` subcategory of the `Input` category.



The V1V2 version reads the sound loudness in the **Sensors** category.

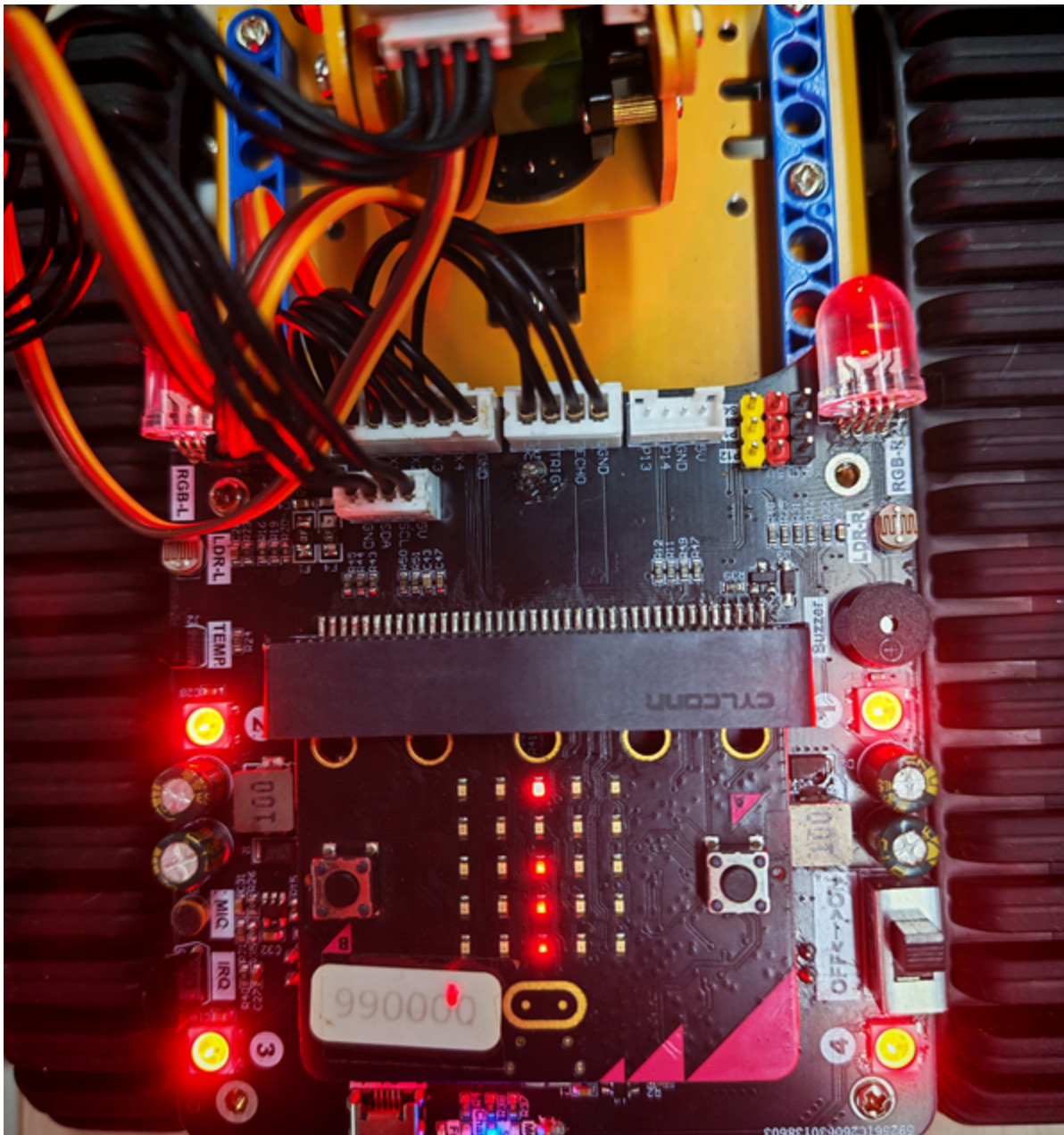


3. Experiment Steps

1. Before flashing, turn off the car switch (switch it to OFF), and connect the Micro USB cable between the computer and the Micro:bit. Note: not the Charging interface on the expansion board.
2. **Micro:bit V2 version** flashes `4.Blowing_Candles_v2.hex`, **Micro:bit V1 version** flashes `4.Blowing_Candles_v1v2.hex`. For the flashing method, refer to section "4. Flashing Programs" in "2. Introduction to Graphic Programming"
3. Turn on the car switch (switch it to ON).

4. Experiment Phenomenon

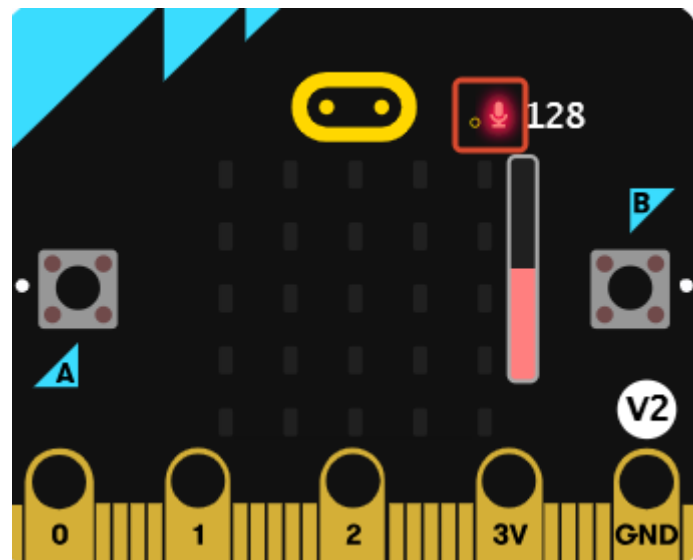
After flashing and turning on the power, the car will first light up the red light, and the LED matrix will show a red candle pattern.



After the red candle pattern disappears, blow air at the microphone. When all the LEDs on the matrix are lit, the car lights turn off and a smiley face is displayed, simulating the candle being blown out.

Blowing air close to the microphone recording position makes it easier to blow out the candle

The recording position of the V2 version is as follows.



The recording position of the V1V2 version is as follows.

